

NIRBHAY GARG

B.Tech CSE (Cyber Security & Digital Forensics) | Backend & Full-Stack Developer

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PROFILE

Second-year B.Tech CSE student focused on backend development, web applications and practical security. Comfortable building APIs, authentication flows and database-backed applications, with hands-on work in Python, FastAPI, Node.js and React.

PROJECTS

Expense Tracker REST API | *Personal Project* | Jul 2026 – Present | [GitHub Repo ↗](#)

- Built a REST API for expense management with FastAPI, SQLAlchemy and PostgreSQL, including Pydantic request validation.
- Added JWT-based authentication and bcrypt password hashing for user accounts.
- Set up Alembic migrations and AWS Boto3 integration; used UV and Ruff during development.

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, Alembic, JWT, Pydantic, Bcrypt, AWS Boto3, UV, Ruff, Git

Lakshya NGO – Full-Stack Website | *Full-Stack Project* | 2026 | *Built and tested, pending launch* | [GitHub Repo ↗](#)

- Designed and built the organization's complete website, including a donation flow with custom amounts and Razorpay payment integration.
- Added UPI / bank-transfer options and server-side validation for donation amount and email inputs.
- Implemented timing-safe HMAC verification and rate limiting on donation endpoints; wrote Vitest tests for frontend and server behavior.

Tech: Node.js, React, Razorpay, HMAC, Vitest, Git

College Attendance Tracker | *Web + Android Personal Project* | 2026

- Built a cross-platform app that signs in to a college student portal, collects attendance and timetable data, and calculates classes that can be skipped against a configurable minimum.
- Created a privacy-focused admin dashboard showing aggregate usage statistics without exposing user credentials or student IDs.

Tech: Web Scraping, Authentication, Web, Android

AI-Based Satellite Image Change Detection | *All India Grand Challenge, IIT Delhi* | Nov 2025

- Built a machine-learning / computer-vision solution to detect changes between multi-temporal satellite images.
- Processed georeferenced imagery and visualized detected changes using Matplotlib and GeoPlotLib.

Tech: Python, NumPy, Rasterio, GeoPlotLib, Matplotlib, Computer Vision, Machine Learning, Git

EXPERIENCE

Intern, Lakshya – A Society for Social & Environmental Development | *SRIJAN Social Internship Programme*

- Carried out CSR fieldwork and documentation for social and environmental development initiatives.
- Designed and built the organization's complete website, including its donation and payment functionality.

EDUCATION

B.Tech, Computer Science & Engineering (Cyber Security & Digital Forensics) | UPES, Dehradun | 2025 – 2029

SKILLS

Languages	Python, C, SQL
Backend	FastAPI, Node.js, REST APIs, SQLAlchemy, Pydantic, JWT Authentication
Frontend	React
Databases	PostgreSQL, SQLite, Alembic
AI / Data	Computer Vision, Image Processing, NumPy, Rasterio, Matplotlib
Cloud & DevOps	AWS (Boto3), Azure VM, DigitalOcean VPS, Cloudflare DNS, Git, GitHub
Security	JWT, bcrypt, HMAC verification, rate limiting, input validation
Tools	UV, Postman, VS Code, Linux, Ruff, Vitest

CERTIFICATIONS

AWS Solutions Architecture Job Simulation | Jan 2026

J.P. Morgan Software Engineering Job Simulation | Oct 2025